

Trickle Down

The 127-year-old theory that fashion flows from the runway to the rack to the mill – **measured**, for the first time.

A research asset • P. Kochar • 2026 •
github.com/pravitkochar/Fashion_Finance

Everyone knows
fashion trickles down.
Nobody had measured how long
it takes.

Veblen sketched it in 1899. Simmel formalized it in 1904. The trade press repeats it every season: a look leaves the runway, lands in Zara a few seasons later, and eventually moves order books at the mills that spin the fiber.

In 127 years, no one published **how
many months, at what strength, for
which materials.**

10-11

MONTHS

A read on the U.S. apparel-demand cycle, roughly a year before the government prints it.

The same fabric shifts that lead the runway lead clothing-store retail-sales turning points by **10-11 months** — earlier than any conventional demand series, built entirely from public data.

Three tiers of the supply chain, aligned on one clock.

26,838

RUNWAY LOOKS

22 houses · 23 seasons · FW2015–FW2026, each machine-tagged for its fabric mix.



11,229

ARCHIVED FABRIC LABELS

267 retailer-months of Zara, H&M, ASOS & Uniqlo rack composition, back to 2016-01.



4+2

DEMAND & PRICE SERIES

Search interest for 22 materials, listed fiber-supplier prices, and apparel retail sales.

Every observation carries the date it became knowable. Nothing looks into its own future.

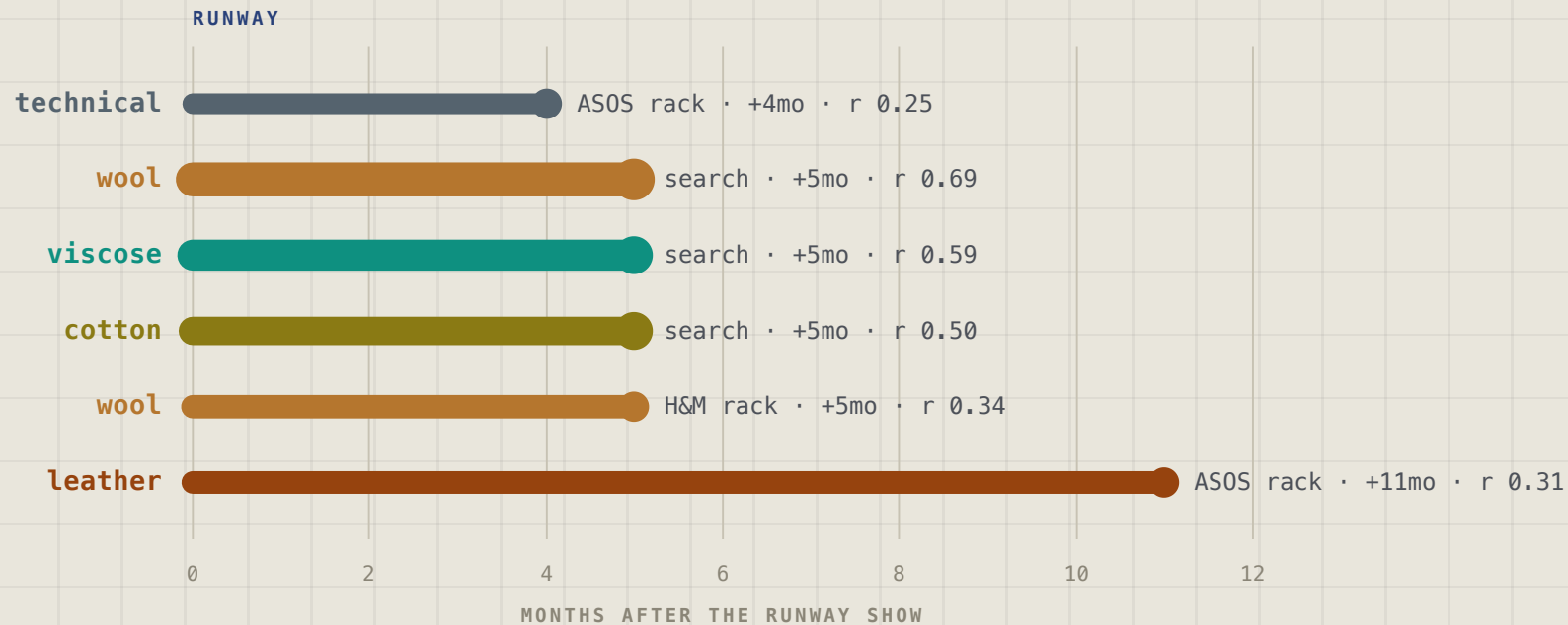
The retailers never kept the history. The Internet Archive did.

Fast-fashion sites overwrite. There is no record of what Zara stocked in 2018 — except that the Internet Archive crawled and froze those product pages, fabric labels and all, with a timestamp on every one.

We read **11,229 composition labels** off archived pages to rebuild 267 months of what the high street actually sold. As far as the literature shows, **nobody has used the Archive this way** — as a point-in-time record of commercial assortment.

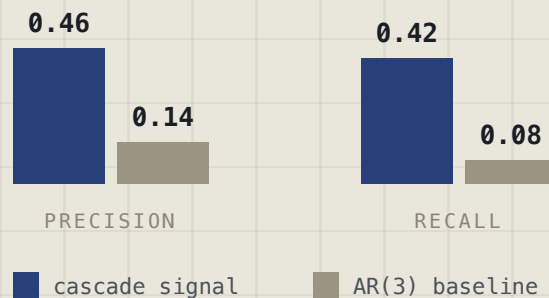
The prior art fits lags of at most two months, between cities or brands. None fits the multi-quarter jump across tiers. This does.

43 fitted links, runway to rack to search.



Fitted on 2017–2022 only, point-in-time. Wool leads consumer search by five months ($r = 0.69$, $n = 78$); the runway leads H&M and ASOS rack space by four to twelve.

It calls apparel turning points before the data does.



Against twelve peaks and troughs in U.S. clothing-store sales growth, the cascade signal leads by 10–11 months at **0.46 precision** and **0.42 recall** – catching 5 of 12.

An AR(3) baseline manages 0.14 and 0.08. Persistence catches nothing.

Drop the four COVID reversals – the easy hits – and it still runs 0.30 / 0.38 on eight turning points.

We tried to trade it three ways. All three failed – under rules we set in advance.



Continuous nowcast

All nine pre-registered parameter sets lost money in walk-forward validation. Best information ratio -0.39 .



Adoption factor

The archive supports depth on only two retailers. A two-name cross-section is a pair trade, not a factor.



Earnings event study

Headline $+3.1\%$ per event – but a pre-event placebo window shows a larger move. The announcement isn't the cause.

We publish the failures at the same size as the wins. That is the point: every number here survived a rule written before we looked. The forecast is what's left standing.

A short list of buyers who already pay for less.

Trend-forecasting vendors

Heuritech, Trendalytics and peers sell shallower runway-signal products at **€12-35k / year** per seat. Ours is measured propagation, not eyeballed.

Textile & fiber planners

Mills and material buyers plan capacity months ahead. A quantified demand lead per fiber maps directly onto their order cycle.

Alternative-data desks

Consumer and macro funds buy anything that leads a print. A reproducible apparel-cycle indicator is exactly that shape.

| Built bottoms-up from named buyers – not a top-down market cone.

The corpus can't be re-bought.

01

A frozen record

The archived assortment history is fixed. A competitor starting today cannot reconstruct what they didn't capture.

02

Deepens every season

Each new fashion week and each monthly crawl extends the panel. The lead time to reproduce it only grows.

03

Method, not luck

The pipeline generalizes to any category whose product pages disclose composition, ingredients or specs.

Three things ship monthly.

The indicator

One number: the runway-weighted material signal, dated, with its historical lead over apparel sales.

Derived signals only. No runway imagery, no scraped retailer content leaves the pipeline.

The lag tables

Per material, per retailer: how many months the runway leads, and how tightly, with confidence.

Emergence alerts

The materials rising fastest off the current runway — wool is up nine points this season — and when they should land.

This was not buildable three years ago.

Vision got cheap

Tagging 26,838 runway looks for fabric mix now costs less than a rounding error. In 2021 it was a research budget.

The archive got deep

A decade of crawled retail pages is finally long enough to fit multi-year lags with real degrees of freedom.

Planning got harder

Demand volatility and inventory risk have made an early, quantified read on material demand worth paying for.

The method is public. The signal is the product.

Every line of the pipeline, the full working paper, and a live dashboard are open at github.com/pravitkochar/Fashion_Finance.

The reproducibility is the credibility — and the frozen corpus behind it is the edge.

Looking for a design partner to turn the indicator into a monthly feed, and the buyers who want the apparel cycle a year early.